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BUMPASS FIRE & RESCUE
FIPS-VALIDATED CRYPTOGRAPHY POLICY

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Owner: Mike Sadler, IT Administrator
Approved By: Chief Randy Bowles

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1. PURPOSE

This policy requires the use of FIPS 140-2 or FIPS 140-3 validated cryptographic modules when protecting the confidentiality of CUI.

2. FIPS MODE REQUIREMENTS

All systems processing CUI must use FIPS-validated cryptography for:

- Encryption of data at rest (BitLocker with FIPS mode)
- Encryption of data in transit (TLS 1.2+ with FIPS-approved cipher suites)
- VPN encryption (AES-256, SHA-256/384)
- Authentication (Kerberos with AES-256)

3. FIPS MODE ENABLEMENT

Windows FIPS compliance mode is enabled via Group Policy:
Computer Config > Windows Settings > Security Settings > Local Policies > Security Options:
"System cryptography: Use FIPS compliant algorithms for encryption, hashing, and signing" = ENABLED

Effect: Windows enforces only FIPS-approved algorithms for TLS (disables RC4, MD5, DES, 3DES-112).

4. APPROVED CRYPTOGRAPHIC ALGORITHMS

Approved for CUI protection:

Symmetric: AES-128, AES-256

Hashing: SHA-256, SHA-384, SHA-512

Asymmetric: RSA-2048+, ECDSA P-256+

Key agreement: ECDH, DH-2048+

Prohibited:

DES, 3DES, RC4, MD5, SHA-1 (for new implementations)

5. HARDWARE

Kingston IronKey USB drives: FIPS 140-2 Level 3 certified.

Windows BitLocker: uses FIPS-validated AES (Windows FIPS mode).

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Approved: Chief Randy Bowles 06/10/2026
Prepared: Mike Sadler 06/08/2026

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